

BEN AMOR Roua

Embedded Electronics & Telecommunications Engineer

[linkedin.com/in/ben-amor-roua](https://www.linkedin.com/in/ben-amor-roua) • github.com/Rouabna

rouabenamor01@gmail.com

Tunis, Tunisia • +216 94 490 729



▸ Profile

Recent graduate engineer in Embedded Electronics and Telecommunications (ESPRIM), specialized in embedded systems, IoT, industrial automation and AI/ML integration. Hands-on IoT development experience (embedded, backend, web/mobile front end) gained through four internships, including a final-year project at Sagemcom on an intelligent automated test bench.

▸ Education

Engineering Degree in Embedded Electronics and Telecommunications 2023 – 2026

École Supérieure Privée d'Ingénieurs de Monastir (ESPRIM), Tunisia – **Class Valedictorian (Major de promotion)**

Bachelor's Degree in Embedded Systems 2020 – 2023

Institut Supérieur d'Informatique et de Mathématiques de Monastir (ISIMM), Tunisia

Tunisian Baccalaureate, Technical Sciences 2016 – 2020

Lycée Ahd Jadid Douz, Kebili, Tunisia

▸ Professional Experience

Embedded Systems & AI Engineering Intern (PFE) – Sagemcom, Tunisia Feb. – June 2026

Design and implementation of an intelligent automated test bench for FXS interfaces on residential gateways, combining an analog measurement chain, real-time monitoring and AI-based anomaly detection, allowing production drifts to be anticipated and costs linked to factory product returns to be reduced.

- Designed the analog measurement chain (relays, signal conditioning) and produced the electronic schematic and PCB layout of the test bench, with simulation and validation in Proteus ISIS before hardware integration.
- Developed embedded control on Raspberry Pi and Basic Stamp 2, with I2C and Telnet communication to the Sagemcom gateway.
- Set up a gateway management server and developed a Flask backend with real-time communication via Socket.IO.
- Developed the web and mobile (iOS/Android) application front end with React Native, including real-time charts for visualizing measurements and test results.
- Integrated an SQLite database for the history of test results and measurements.
- Implemented an AI anomaly-detection module (Isolation Forest).

Raspberry Pi Basic Stamp 2 Python Flask Socket.IO React Native SQLite Proteus ISIS ADS1115 (ADC)
INA219 (power) AD736 (AC-DC conv.) PCB Design I2C Telnet Isolation Forest Machine Learning

Industrial IoT Developer Intern – Sartex, Tunisia July – Sept. 2025

Development of an industrial IoT monitoring system for real-time tracking of textile water basins, integrating Delta PLCs and a connected web/mobile application.

- Designed and implemented a Node.js backend handling Modbus TCP communication with three Delta PLCs.
- Programmed the three PLCs in ISPSOft, adding an HMI command for bidirectional control via Modbus TCP.
- Developed a React Native web and mobile interface for remote monitoring and control.
- Integrated an SQLite database and set up real-time communication via WebSocket.
- Carried out functional testing and validated communication between the PLCs and the application.

Node.js express.js React Native Modbus TCP Delta PLCs ISPSOft COMMGR SQLite WebSocket REST API
HTTP

Electronics Technician Intern – SE Engineering SARL, Tunisia Feb. – June 2023

Design of a complete IoT system for real-time measurement, monitoring and control of energy consumption, based on an ESP32 microcontroller and a Flutter application dedicated to display, history and alert management.

- Developed an embedded prototype based on ESP32 with current and voltage sensors.
- Designed a Flutter application for real-time display and visualization of consumption history.
- Implemented an automatic alert system in the event of a power outage or anomaly.
- Produced the electronic schematic and PCB layout in Altium Designer.

Flutter Arduino IDE Firebase Google Sheets Altium Designer Fritzing Dart Sensors

Introduction to the telecommunications field and network infrastructure within Tunisie Telecom's technical department.

- Observed the operation of data centers and network distribution points.
 - Learned about transmission and switching equipment and communication protocols.
-

▷ Technical Skills

- **Languages:** C, C++, Python, JavaScript, Java, Dart, VHDL
 - **Embedded Systems:** Arduino, STM32 (CubeIDE/MX, FreeRTOS), ESP32, Raspberry Pi, FPGA
 - **Web & Mobile:** Node.js, React Native, Flutter, HTML/CSS, Express.js
 - **AI / Machine Learning:** TensorFlow, Keras, OpenCV, CNN, Isolation Forest
 - **Communication & IoT:** MQTT, WebSocket, REST API, Modbus TCP, I2C, SPI, UART, Bluetooth
 - **PCB & CAD:** Altium Designer, Proteus ISIS
 - **Databases:** SQLite, MongoDB, Firebase
 - **Tools & Methods:** Git/GitHub, Docker, Jira, Scrum, VS Code
-

▷ Academic Projects

- **Autonomous AI Defense Turret** – Developed an embedded system capable of detecting human presence and reacting in real time with automated defense actions, including deep-learning models for facial recognition using a PiCamera. (*Python, TensorFlow, OpenCV, Raspberry Pi, SolidWorks, SSH*)
 - **Smart Road Sign Detection** – Designed a real-time system for detecting and classifying road signs, using a PiCamera for image capture and image-processing algorithms for shape and color recognition. (*OpenCV, NumPy, Raspberry Pi, Python*)
 - **Smart Home Automation (MQTT)** – Built a home automation solution based on the MQTT protocol to automate lighting and connected devices, with a Node.js server, a MongoDB database, and an interactive web dashboard for remote control and monitoring. (*Node.js, MQTT/Aedes, Express.js, MongoDB*)
 - **CNN Image Classifier** – Created a binary image classifier using a convolutional neural network (CNN) with TensorFlow, providing real-time predictions on image streams. (*TensorFlow, Python*)
 - **IoT Health Monitoring** – Developed a system to track a patient's vital signs via ESP32: sensor data collection, transmission to a web server, real-time visualization, and automatic alerts for abnormal values. (*ESP32, C/C++, Sensors*)
-

▷ Certifications

- AI for Predictive Maintenance – NVIDIA Deep Learning Institute
 - Industry 4.0 Fundamentals
 - STM32 Microcontroller Programming
-

▷ Languages

Arabic (native) • French (advanced) • English (advanced) • Spanish (beginner)

▷ Personal Skills

Teamwork • Complex Problem Solving • Adaptability • Autonomy • Time Management